
BY 524P: Medical and NanoBiotechnology Lab

Course Code	BY 524P
Course Name	Medical and NanoBiotechnology Lab
L-T-P-C	0-0-2-1
Prerequisites	M.Tech. Biotechnology Students who enrolled for Medical and NanoBiotechnology specialization
Intended for	M. Tech. Biotechnology
Distribution	Core Lab for M. Tech. Biotechnology
Approval	9th Senate (May 2016)

Course Contents

Cellular Fuel and Cellular Communication

- Glucose production assay from hepatocytes cells
- Determine the effect of fasting and feeding in glucose output from hepatocytes cells
- Hormonal regulation of gluconeogenic key regulators at their transcripts (real time PCR) and protein (Western blotting) levels
- Determine the effect of Insulin and Glucagon in regulating various second messengers (cAMP, Ca²⁺) level and their downstream effector molecules from hepatocytes cells

Disease Biology:

- Culture and Gram's staining of Gram positive and negative bacteria,
- Isolation and characterization of some common pathogenic bacteria from human samples.
- ELISA for viral diseases (hepatitis and measles) and/or smRNP
- EITB for parasitic disease (cysticercosis) diagnosis.
- Immunofluorescence for lupus diagnosis
- Enzyme assay for insulin activity from hepatocyte cell line.

NanoBiotechnology:

- Synthesis and characterisation of gold nanoparticles.
- Synthesis and characterization of silver nanoparticle.
- Synthesis and characterization of quantum dots.
- Understanding the analyte specific aggregation of nanoparticle for biosensing application.
- Studying the antibacterial property of silver nanoparticles.
- Synthesis and characterization of polymeric nanoparticles as a drug delivery vehicle.
- Fabrication of nanoparticle impregnated antibacterial polymeric films.

Protein Sciences in Therapeutics:

- Site-directed mutagenesis
- Protein structure modification
- Visualization of proteins for engineering purpose
- Obtaining highest purity of Proteins using gel filtration
- Ion Exchange methods to purify proteins
- Biophysical characterization of proteins.
- Protein folding in-vitro
- pH and salt dependent studies of proteins.
- Protein aggregation